

Real Credit Cycles[†]

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We embed diagnostic expectations in a workhorse neoclassical model with heterogeneous firms and risky debt. A realistic degree of overreaction estimated from US firms' earnings forecasts generates realistic credit cycles. Good times produce economic and financial fragility, predicting future disappointment of expectations, low bond returns, and investment declines. To generate the size of spread increases observed during 2007–2009, the model requires only moderate negative shocks. Diagnostic expectations offer a realistic, parsimonious way to produce financial reversals in business cycle models. (JEL D84, E13, E22, E32, E44, G12, G32)

Developed economies experience boom-and-bust cycles in real and financial activity. Booms in which credit spreads are low and investment, output, and leverage are high predict busts in which spreads rise and investment, output, and leverage decline (Schularick and Taylor 2012; López-Salido, Stein, and Zakrajšek 2017), even absent bad fundamental shocks. In most accounts, busts are due to “financial” shocks, such as a spike in investors’ required returns (Korinek and Simsek 2016), uncertainty (Arellano, Bai, and Kehoe 2019), or a bank run (Diamond and Dybvig 1983), which are amplified by fire sales or demand externalities (Bernanke and Gertler 1989; Kiyotaki and Moore 1997; Eggertsson and Krugman 2012). But what are financial shocks? And why do they occur after booms? Existing theories do not answer these questions.¹

We offer an alternative approach in which boom-and-bust cycles are systematic due to nonrational expectations. We microfound previous informal arguments

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¹ Shleifer and Vishny (1992) offer the original treatment of fire sales. This amplification is also in Lorenzoni (2008); Stein (2012); and Dávila and Korinek (2017). Demand externalities are studied by Farhi and Werning (2020); Guerrieri and Lorenzoni (2017); Korinek and Simsek (2016); and Rognlie, Shleifer and Simsek (2018).

(Minsky 1977; Kindleberger 1978) by embedding diagnostic expectations (DE) (Bordalo, Gennaioli, and Shleifer 2018) in a workhorse neoclassical model with defaultable debt. DE introduce a single new parameter capturing belief overreaction to total factor productivity (TFP) news. We discipline beliefs using data on managers' expectations about their firms' profitability.

A realistic degree of overreaction produces realistic credit cycles. During good times, investors are too optimistic, so credit and investment overexpand. When beliefs cool off, credit markets tighten, real activity declines, and default rates rise, consistent with Bordalo, Gennaioli, and Shleifer (2018) and Bordalo et al. (2019). The model yields sharp reversals even in the absence of large negative TFP shocks and matches the nontargeted predictability of realized bond returns at the firm and aggregate levels, without assuming exotic risk preferences. We show that under rational expectations, financial frictions amplify shocks but cannot generate meaningful countercyclical volatility of credit spreads, in particular realistic spread hikes and drops in debt and investment after small negative TFP shocks.

In our setting DE give rise to the following expectations formula, founded in the psychology of selective recall:

$$(1) \quad \mathbb{E}_t^\theta(A_{t+1}) = \mathbb{E}_t(A_{t+1}) + \theta[\mathbb{E}_t(A_{t+1}) - \mathbb{E}_{t-1}(A_{t+1})],$$

where A_{t+1} is future TFP, $\mathbb{E}_t(\cdot)$ is the rational expectation at time t , and $\theta \geq 0$ is a diagnosticity parameter. When $\theta > 0$ agents overreact because good news does not just increase the true likelihood of good future outcomes, it also causes such outcomes to be top of mind and thus overweighted in beliefs. DE have two advantages relative to models of adaptive expectations (Cagan 1956) or of exaggerated TFP persistence (Angeletos, Huo, and Sastry 2020; Greenwood, Hanson, and Jin forthcoming). First, belief distortions depend on the data-generating process experienced by the agent, making DE immune to the Lucas (1976) critique. Second, under DE, optimism swiftly cools off when good news recedes into the past, even in the absence of bad shocks. As a result, small shocks cause significant instability, which does not arise when—for instance—agents merely exaggerate persistence. Third, and importantly, DE modify RE by a single parameter, θ , that has been estimated in different datasets, including the forecasts of financial (Bordalo et al. 2019) and macroeconomic analysts (Bordalo et al. 2020). These estimates discipline our quantifications.

The paper proceeds as follows. In Section I we present novel evidence on boom-and-bust cycles in expectations, bond returns, and investment at the firm level. We show that managers' expectations of their firms' profits overreact: They are too optimistic in good times and too pessimistic in bad times. In turn, excess optimism predicts a one -year-ahead decline in the firm's bond return as well as lower investment growth. We then investigate whether these micro-level boom-and-bust dynamics (i) can be explained by a disciplined departure from rationality and whether (ii) they can produce aggregate instability.

Section II introduces the DE model, discussing its foundations and implications. Sections III and IV introduce DE about firms' TFP in a model in which consumers are risk neutral and firms hire labor, invest, and issue equity as well as risky debt subject to idiosyncratic and aggregate TFP shocks. In our main analysis the wage and the interest rate are fixed (deep-pocketed lenders provide credit), but we endogenize an

equilibrium wage in Section VIB. We structurally estimate the model by matching firm-level moments on profitability, spreads, debt, investment, and forecast errors. The estimated degree of diagnosticity $\theta \approx 1$ is in the ballpark of previous estimates (Bordalo et al. 2019; Pflueger, Siriwardane, and Sunderam 2020).

The model yields two key results. First, under DE, but not under an estimated RE version of the model, the reaction of aggregate investment to TFP shocks is highly nonlinear. In good times, even a small negative TFP shock causes a large drop in aggregate investment, while the same shock has a muted effect in normal times. Second, in the DE model large increases in credit spreads occur even after only moderately negative TFP shocks. Under RE, by contrast, spreads rise little even after much larger negative TFP shocks. Due to their overreaction and sudden reversal, DE offer a theory of amplification through “financial shocks”: Waning of excess optimism causes a sudden inward shift in the supply of capital, causing rising credit spreads (Jermann and Quadrini 2012; Gilchrist and Zakrajšek 2012).

In Section V we assess the model’s ability to match untargeted facts. We first look at macro comovements. The DE model overall outperforms the RE model among key financial dimensions, producing volatile and countercyclical credit spreads. These are central features of credit cycles, and they are again due to the boom-and-bust nature of DE. Overreaction produces an outward shift in the supply of capital during good times, followed by an inward shift due to belief reversal. In a canonical business cycle accounting exercise (Chari, Kehoe, and McGrattan 2007), we show that belief fluctuations manifest as a countercyclical investment wedge, which drops during periods of overoptimism and rises during periods of overpessimism.

We then go back to the firm-level boom-and-bust cycles in expectations, investment, and bond returns documented in Section I. We show that the DE (but not the RE) model offers a good account of them. We also show that the estimated DE model can produce realistic cycles at the sectoral level, in which good times are followed by disappointment and lower credit spreads. These patterns are also fully untargeted in our calibration.

As a final exercise, we ask how large a TFP shock is required in the DE model to produce the massive increase in credit spreads observed during 2007–2009 and investigate the macroeconomic implications of such a shock. We find that a moderate decline in TFP is sufficient to generate the observed increase in credit spreads. The RE model is entirely incapable of matching this performance, with little movement in spreads in response to the identical shock. The DE model also produces larger declines in aggregate investment and earnings expectations, explaining a higher fraction of the observed declines in each variable.

Section VI performs two robustness exercises. First, we endogenize the real wage in general equilibrium by adding a household labor choice with convex effort cost. As expected, this modification dampens volatility but does not eliminate sizable fragility after good times, even with flexible wages. Second, we perform sensitivity analysis by varying a range of model parameters. Again, our key results appear to be robust.

Our paper contributes to two strands of work in macroeconomics. Work on macrofinancial instability maintains rational expectations but produces time-varying risk attitudes through habits (Campbell and Cochrane 1999), long-run risk (Bansal and Yaron 2004), or disaster risk (Barro 2006). In these accounts, time-varying risk premia are not directly measured, and predictions regarding survey expectations

are counterfactual. Habit formation predicts countercyclical expected returns, while survey expectations of returns are procyclical (Greenwood and Shleifer 2014). Investors expect low returns when they expect a crash, contrary to the predictions of disaster risk models (Giglio et al. 2021). Here, we assume standard preferences, in particular constant required returns, but relax belief rationality in a manner disciplined by micro data. Future work may combine the two ingredients.

A second strand of work studies departures from full information rational expectations due to rational inattention (Sims 2003; Mankiw and Reis 2002). Coibion and Gorodnichenko (2015) study inflation forecasts using information rigidities as in Woodford (2003). Kohlhas and Walther (2021) model asymmetric attention paid to distinct macroeconomic variables. Kozłowski, Veldkamp, and Venkateswaran (2020) links belief dynamics to the persistence of the Great Recession. This work does not produce excess volatility in beliefs and economic outcomes. Angeletos, Huo, and Sastry (2020) combine dispersed information and exaggerated persistence. We show that exaggerated persistence cannot produce quantitatively realistic belief volatility. A different approach emphasizes ambiguity aversion (Hansen and Sargent 2001; Ilut and Schneider 2014; Bianchi, Ilut, and Schneider 2018; Bhandari, Borovicka, and Ho 2025), producing excess volatility in beliefs, but based on an underlying pessimism that is difficult to square with observed excess optimism in beliefs data.² Adam, Marcet, and Beutel (2017) and Farmer, Nakamura, and Steinsson (2024) model expectations that may deviate from rationality for sustained periods due to agents learning about hard-to-identify processes based on misspecified models. Here, we estimate overreaction also using purely idiosyncratic firm-level TFP shocks, which are arguably not tied to hard-to-identify aggregate conditions.

Our key innovation is to discipline belief overreaction using micro data and study its macrofinancial implications. Recent work introduces DE into neoclassical and New Keynesian models (Bianchi, Ilut, and Saijo 2023; L'Huillier, Singh, and Yoo 2024) and shows that they generate quantitatively meaningful amplification and propagation of traditional business cycle shocks. Relative to this work building DE into canonical representative agent business cycle models, we add both financial frictions allowing for a study of credit spreads and a heterogeneous firms model allowing for estimation based on microdata. Further building out our model to include additional business cycle mechanisms such as aggregate demand shifts (Farhi and Werning 2020), time-varying capacity utilization (King and Rebelo 1999), or sharp cuts to firm-level employment in bad times (Ilut, Kehrig, and Schneider 2018) may offer an even more realistic model of macroeconomic volatility. Maxted (2023) introduces DE in He and Krishnamurthy's (2019) model of bank runs. Krishnamurthy and Li (2025) also study beliefs and intermediation in a DE framework. Falato and Xiao (2024) studies overreaction in bank expectations. Chodorow-Reich, Guren, and McQuade (2024) studies overreaction in housing price expectations. Overreacting beliefs about future earnings growth can account for longstanding stock market puzzles (Bordalo et al. 2024a), including return predictability, and shed light on

²See also Falato and Xiao (2020) and Schaal and Taschereau-Dumouchel (2023) on dispersed information. Jaimovich and Rebelo (2007) study the impact of other belief distortions in macro models without financial frictions. Caballero and Simsek (2020) study demand externalities in models with extrapolative beliefs. Behavioral finance has studied credit cycles (Greenwood, Hanson, and Jin forthcoming), including under DE (Bordalo, Gennaioli, and Shleifer 2018), without, however, assessing the quantitative performance of these accounts.

macrofinancial fluctuations (Bordalo et al. 2024b), even in the absence of time variation in required returns.

I. Data

We use survey data to connect belief overreaction to boom-and-bust cycles in investment and bond returns at the firm level, controlling for macro shocks (S&P Global and CRSP 2020; Thomson Reuters 2017, 2019; WRDS 2020b,a; WRDS, Mergent, and FINRA 2020; BEA 2020b,a; US Treasury 2020; Fernald 2020).

We use micro data on firm-level forecasts from the IBES manager guidance database, which records—for an individual firm-fiscal year—a manager’s forecast for their company’s profits or earnings over the next year. We exploit forecasts made concurrently with the release of the year’s financials, in a sample spanning the 1999–2018 period. We link these data to Compustat, which provides standard financial information. We also use the Mergent Fixed Income Securities Database (FISD), which contains issuance information including spreads on individual securities, and the FINRA’s Trade Reporting and Compliance Engine (TRACE) dataset, which contains data on secondary market transactions from bond dealers and hence allows us to compute realized bond returns. The FISD-TRACE sample covers the years 2003–2018. Supplemental Appendix B provides more information on data sources and sample construction, as well as descriptive statistics.

To assess whether expectations overreact, we regress a firm’s next-year forecast errors, defined as realized minus predicted profits, on current-year firm-level profits, investment, debt issuance, and profit forecasts, controlling for time and firm effects. Under rational expectations, the manager’s forecast errors should be unpredictable based on any information available to them when the forecast is made. If beliefs overreact, displaying overoptimism during good times and pessimism during bad times, then future forecast errors should be negatively correlated with the firm’s current fundamentals, proxied by investment and debt issuance. Table 1 reports results consistent with overreaction: Firms earning more, investing more, issuing more debt, or forecasting higher future profits on average experience more negative earnings surprises in the future.

This evidence confirms the violations of rationality found by Gennaioli, Ma, and Shleifer (2016) and Barrero (2022) in different datasets and contexts. Overreaction is a robust feature of firm forecasts across private and publicly listed firms, with differing strategic incentives (some forecasts are confidential, others are publicly disclosed).³ The magnitudes are meaningful. In column 2 a 1 standard deviation (0.23) higher investment rate on average predicts $0.457 \times 0.23 \approx 10.5$ percentage points stronger disappointment in earnings next year.

These results are robust to alternative specifications, as reported in Supplemental Appendix Table B.3. We exclude the Great Recession, to avoid the influence of outliers, exclude firms with less than five years of earnings guidance (which are more

³Recent work documents overreaction by correlating forecast errors with forecast revisions (Bordalo et al. 2020, 2019; Coibion and Gorodnichenko 2015). Here, we do not have enough data to perform this type of revision-based analysis. Bouchaud et al. (2019) study equity analysts’ forecasts. They document that these forecasts, while correlated with managerial forecasts, display a form of underreaction.

TABLE 1—PREDICTABLE FORECAST ERRORS

Dependent variable:	Forecast error _{t+1}			
Estimation method:	OLS (1)	OLS (2)	OLS (3)	OLS (4)
Profits _t	−0.044 (0.021)			
Investment _t		−0.457 (0.065)		
Debt _t			−0.326 (0.063)	
Forecast _t				−0.244 (0.031)
Firm effects	✓	✓	✓	✓
Year effects	✓	✓	✓	✓
Years	1999–2018	1999–2018	1999–2018	1999–2018
Firm-years	9,666	9,666	9,666	9,666

Notes: The table reports panel OLS estimates from the merged Compustat-IBES sample of the coefficients of a regression of forecast errors on the indicated variable. The standard errors are clustered at the firm level. All variables are measured at the firm-fiscal year level. Forecast errors in $t + 1$ are realized earnings in $t + 1$ minus firm forecasts in t , scaled by firm tangible assets. Investment in t is capital expenditures, scaled by firm tangible assets. Debt is long-term and short-term liabilities at the end of t , scaled by firm total assets. Forecasts are manager guidance in the year t earnings call about earnings in year $t + 1$, scaled by firm tangible assets. Note that 0.01 = 1% relative to the firm's assets.

likely to manipulate earnings forecasts (Bertomeu et al. 2022)), and exclude firms with high-yield debt, to verify that overreaction is not restricted to risky firms. We continue to uncover evidence of forecast overreaction. We also show that the predictability of forecast errors works symmetrically in good and in bad times, defined as periods with above- or below-average investment or debt.

These patterns also do not reflect simple constant heterogeneity in optimism across firms since each column controls for firm fixed effects. But one might be concerned that firm fixed effects generate the downward dynamic panel bias of Nickell (1981) in finite samples.⁴ So in Supplemental Appendix B, we develop a GMM estimator based on covariances of the *growth* of outcomes within firms up to the current period with future forecast error growth. The estimator is asymptotically equivalent to the OLS specifications in Table 1 but not subject to the same dynamic panel bias. As Supplemental Appendix Table B.2 shows, our baseline overreaction patterns go through and, if anything, strengthen in magnitude using this alternative estimator.

If overreaction produces boom-and-bust cycles, belief disappointment should be correlated with reversals in firm-level outcomes. To assess this prediction, Table 2 performs a two-stage regression exercise. In the first stage (see column 1), we regress future forecast errors on current investment, analogous to Table 1 for this

⁴Intuitively, the forecast errors on the left-hand side of these regressions include profit innovations, which in finite samples may influence the mean forecast errors subtracted by the OLS fixed effects estimator. If firm financial outcomes on the right-hand side are correlated with these profit innovations, there will be a mechanical negative correlation resulting in downward bias of the OLS fixed effect estimator.

TABLE 2—LINKING FORECAST ERRORS AND FIRM REVERSALS

Dependent variable:	Forecast error _{<i>t</i>+1}	Return _{<i>t</i>+1}	Δ Investment _{<i>t</i>+1}
IV stage:	First (1)	Second (2)	Second (3)
Forecast error _{<i>t</i>+1}		0.007 (0.004)	0.483 (0.061)
Investment _{<i>t</i>}	−0.565 (0.104)		
Year effects	✓	✓	✓
Profit control			✓
Years	2002–2018	2002–2018	2002–2018
Firm-years	2,852	2,852	2,852
First-stage <i>F</i>	29		

Notes: The table reports estimates of specifications on the merged Compustat-IBES-FISD/TRACE sample at the firm–fiscal year level. Column 1 reports IV first-stage estimates, and the remaining columns report IV second-stage estimates. Column 3 controls for current profits in the second stage. Standard errors are clustered at the firm level. Note that $0.01 = 1\%$ relative to the firm’s assets.

sample, which detects excess optimism (pessimism) as predictably negative (positive) forecast errors.

In the second stage, reported in columns 2 and 3, we regress future bond returns and investment growth on predicted forecast errors, controlling for time effects. Belief-driven cycles imply that overoptimism (negative predicted errors) and overpessimism (positive predicted errors) are followed by reversals in bond returns and investment growth.⁵ To assuage the concern that investment and belief errors may be jointly driven by mean reversion in fundamentals, the investment regression controls for current profits.⁶ The results in column 2 suggest that departures from rationality arise not only for managers (as shown in column 1 and in Table 1) but also for bond market investors, and those in column 3 show that they can produce investment cycles. Later, we confirm these findings using simulated data from our model, which is immune to omitted variables bias.

The results are consistent with belief-driven boom-and-bust cycles. Good times with high investment are systematically followed by disappointment (column 1), low future bond returns (column 2), and low future investment growth (column 3). The opposite occurs after bad times. In column 2 a firm predictably disappointed with a 1 standard deviation (0.78) lower forecast error sees its bond returns fall by $0.007 \times 0.78 \approx 0.5$ percentage points on average. Models without overoptimistic pricing of debt on the part of investors cannot generate this sort of micro-level return predictability, regardless of the belief structure at work within firms. Column 3 reveals that the investment rate declines by $0.483 \times 0.78 \approx 37.8$ percentage points at the same

⁵In Supplemental Appendix B we also check that these firm-level cycles are not driven by risky firms, which may display more cyclicality. We repeat the exercise in Supplemental Appendix Table B.5 for the subset of investment grade firms, excluding all firms with high-yield debt. Our results are confirmed.

⁶Column 1, like Table 1, provides evidence of nonrationality of managers’ profit expectations, a result that is also robust to removing the control for current profits. See Supplemental Appendix Table B.6.

firm. The rest of the paper draws out the aggregate consequences of these firm-level patterns by introducing diagnostic expectations into a standard neoclassical model.

II. Diagnostic Expectations and Neglected Risk

A. Diagnostic Expectations

There are multiple models of belief overreaction in the literature. However, the DE model accounts for a broad range of well-documented departures from rationality, starting from the work of Kahneman and Tversky in the 1970s. DE were developed by Bordalo, Gennaioli, and Shleifer (2018), building on Gennaioli and Shleifer (2010). To see how they work, consider an agent forecasting a future variable X_{t+1} , say TFP, on the basis of its history. The true distribution is Markovian, denoted by $f(X_{t+1}|X_t)$. Under DE, the agent's beliefs follow the distorted distribution:

$$(2) \quad f^\theta(X_{t+1}|X_t) \propto f(X_{t+1}|X_t) \left[\frac{f(X_{t+1}|X_t)}{f(X_{t+1}|\mathbb{E}_{t-J}(X_t))} \right]^\theta,$$

where $\theta \geq 0$ and $\mathbb{E}_{t-J}(X_t)$ is the rational expectation of X_t conditional on information at $t - J$.⁷ In equation (2) beliefs overweight future outcomes that have become more likely on the basis of the last J periods' news. Overweighting is captured by the likelihood ratio on the right-hand side, which measures the current increase in the probability of X_{t+1} relative to the case of J periods of neutral news, $X_t = \mathbb{E}_{t-J}(X_t)$. θ captures the strength of overweighting. When $\theta = 0$, expectations are rational. When $\theta > 0$, beliefs overreact.

Equation (2) is founded on selective recall. The data-generating process, captured by $f(X_{t+1}|X_n)$, $n \leq t$, is stored in the agent's memory database. The agent selectively recalls outcomes that historically became more likely after the most recent J news and fails to recall other outcomes. Thus, beliefs overweight the future outcomes that news cause to be top of mind. Bordalo, Coffman et al. (2021) offer a memory foundation for equation (2) and find support for it in two experiments. See also Bordalo et al. (2023) for related evidence.

To see the implications of equation (2), suppose that TFP follows an AR(1) process, $\ln A_t = \rho \ln A_{t-1} + \varepsilon_t$, where $\rho \in (0, 1)$ and ε_t is Gaussian with mean zero and variance σ^2 . At time t , the diagnostic distribution of next-period productivity is also Gaussian, with mean

$$(3) \quad \mathbb{E}_t^\theta(\ln A_{t+1}) = \rho \ln A_t + \theta \rho \sum_{r=0}^{J-1} \rho^r \varepsilon_{t-r}$$

⁷Equation (2) assumes that X_t is perfectly observable. Bordalo et al. (2019, 2020, 2023) allow for information frictions such as dispersed information about X_t , in which case DE yield a distorted Kalman Filter that reconciles overreaction of individual forecasts with consensus underreaction. In equation (2), as we later discuss, the agent's reliance on the true data-generating process, and hence the dependence of overreacting beliefs on the "rational expectation," is a by-product of selective memory retrieval from an unbiased memory database, not of the agent's awareness of the rational expectation.

and variance σ^2 . DE are too optimistic after J periods of on average good news and too pessimistic after J periods of on average bad news. Future forecast errors are negatively predicted by current TFP:

$$\text{cov}[\ln A_{t+1} - \mathbb{E}_t^\theta(\ln A_{t+1}), \ln A_t] \propto -\theta \rho \sigma^2 \left(\frac{1 - \rho^{2J}}{1 - \rho^2} \right) < 0,$$

which allows DE to account for the error predictability in Table 1. This cannot occur under RE, with $\theta = 0$, which implies a zero covariance above.

We allow overreaction to current news only; that is, $J = 1$. This assumption is common in applications of DE (Bordalo, Gennaioli, and Shleifer 2018), and we adopt it to minimize degrees of freedom. Some applications allow for $J > 1$, showing that in this case DE produce slowly brewing asset price bubbles (Bordalo et al. 2019; Maxted 2023; Bordalo, Gennaioli et al. 2021). In macroeconomics, Bianchi, Ilut, and Saijo (2023) show that longer streaks of overreaction can generate hump-shaped dynamics in New Keynesian models.⁸

Some of our results could arise from alternative models of overreaction. However, we think that DE offer two important advantages. First, they are based on the psychology of memory (Kahana 2012; Bordalo et al. 2023), which gives DE substantial external relevance as well as sharp testable predictions based on this memory structure and the reaction of memory to cues. For example, the role of memory in this context implies that belief distortions are stronger when the underlying persistence ρ is higher, consistent with Bordalo et al. (2019, 2020); Afrouzi et al. (2023); Azeredo da Silveira, Sung, and Woodford (2020). As a result of these memory-based microfoundations, DE are immune to the Lucas critique (Lucas 1976). Reliance of memory on cues creates sharp belief reversals, in which otherwise stable economic conditions cause overoptimism to quickly wane, causing excess volatility. These features are not generally present in models in which agents hold an exogenously misspecified model or follow adaptive learning.^{9,10}

The second key advantage of DE is that deviations from rationality are determined by a single parameter, the degree of diagnosticity θ . θ has been estimated in various settings, ranging from financial analyst firm-level earnings forecasts (Bordalo et al. 2019), to professional forecasters who predict macroeconomic variables (Bordalo et al. 2020), to volatility in stock prices (Pflueger, Siriwardane, and Sunderam 2020). These estimates point to a value of θ between 0.5 and 1.5. The

⁸ Supplemental Appendix Table B.4 provides empirical evidence on the persistence of forecast overreaction up to lag $J = 2$ but finds no effect at lags $J > 2$. It is thus realistic to focus on $J = 1$ in our context.

⁹ Suppose, for instance, that the agent inflates TFP persistence as $\ln A_t = \hat{\rho} \ln A_{t-1} + \varepsilon_t$, where $\hat{\rho} > \rho$, as proposed by Greenwood and Hanson (2013) and Angeletos, Huo, and Sastry (2020). Parameter $\hat{\rho}$ is not linked to the true DGP, violating forward-looking behavior. In addition, beliefs stay overly optimistic as long as TFP is above its long-run mean of $\ln A = 0$, so there are no sharp reversals absent bad news. Substantively, then, realistic belief volatility requires counterfactually high excess persistence. As we show in Supplemental Appendix Section B.6, to obtain belief estimates from Section IV to this model requires $\hat{\rho} > 1$, which is incompatible with dynamic optimization. Similar problems arise in adaptive models of mechanical extrapolation. Another approach to capture extrapolation is Fuster, Laibson, and Mendel's (2010) Natural Expectations.

¹⁰ Kohlhas and Walther (2021) show that overreaction may be produced by overconfidence, which cannot, however, yield overreaction of consensus forecasts, which is documented in Bordalo, Gennaioli, and Shleifer (2022). Kohlhas and Walther (2021) show that overreaction can obtain under a form of rational inattention but not at the level of individual forecasts as instead documented in Bordalo et al. (2020).

similarity of these estimates is noteworthy because it highlights a basic psychological force toward overreaction operating both for idiosyncratic (i.e., firm-level) and aggregate (i.e., macroeconomic) shocks, which we later confirm in our setting. The ballpark of existing estimates of θ provides a valuable external benchmark to discipline our quantitative exercise and assess its reliability.

In sum, although our analysis does not preclude the use of different models, we view equation (3) as a microfounded, tractable, and disciplined way to capture the observed belief overreaction in managers' beliefs in Section I. We next use this model to study our key object of interest: the genesis of financial reversals, such as sudden increases in credit spreads. To do so, we incorporate DE into an incomplete market setting with defaultable debt. We next present a stylized example to illustrate how in such a setting DE affect the credit spread demanded by capital supplier. We also discuss the differences between our approach and models of excess volatility based on uncertainty shocks and time-varying risk aversion.

B. Diagnostic Expectations, Belief Volatility, and Supply-Driven Bond Pricing: A Toy Model

A firm seeks to roll over one-period defaultable bonds. Deep-pocketed, risk-neutral lenders demand a constant expected return $R = 1$. If the probability of default perceived by lenders for the next period is δ_t^θ , the interest rate is $\hat{R}_t = \frac{1}{1 - \delta_t^\theta}$, so the credit spread is

$$(4) \quad \hat{R}_t - 1 \equiv S_t = \frac{\delta_t^\theta}{1 - \delta_t^\theta},$$

which increases in the default risk δ_t^θ perceived by lenders.

Suppose that the firm defaults at time t if and only if its productivity is lower than a threshold, $A_t < A^*$. The perceived probability of default is then given by

$$(5) \quad \delta_t^\theta = \Phi \left[\frac{\ln A^* - \mathbb{E}_t^\theta(\ln A_{t+1})}{\sigma} \right],$$

where $\Phi(\cdot)$ is the standardized Gaussian CDF. Due to DE, perceived risk is too low after good news and too high after bad news.

Equation (5) has implications for the path of credit spreads. Substituting δ_t^θ in (4) and linearizing with respect to $\mathbb{E}_t^\theta(\ln A_{t+1})$ around its long-run mean yields

$$(6) \quad S_t \approx S_\infty - S \mathbb{E}_t^\theta(\ln A_{t+1}),$$

where $S_\infty > 0$ is the long-run spread and S is a positive scalar. The spread falls when creditors are more optimistic about future productivity. Inserting (3) into (6), we obtain

$$(7) \quad S_t \approx S_\infty (1 - \rho) + \rho S_{t-1} - S_\rho (1 + \theta) \varepsilon_t + S \theta \rho^2 \varepsilon_{t-1}.$$

Under RE, with $\theta = 0$, spreads mirror TFP and follow an AR(1) process with persistence ρ . Under DE, with $\theta > 0$, there are two differences. First, current TFP shocks are amplified, as captured by the term $-S\rho(1 + \theta)\varepsilon_t$. This is a source of excess volatility: After a positive TFP shock, beliefs become too optimistic, and the spread declines too much. Second, there is reversal of past TFP shocks, as captured by the term $S\theta\rho^2\varepsilon_{t-1}$. This is a source of boom-and-bust dynamics: In the future, current optimism wanes, and the spread increases.

Due to the second mechanism, DE generate predictable bond returns, which arise because the current spread S_t can differ from its RE counterpart S_t^* on the basis of current shocks. Equation (7) implies

$$(8) \quad S_t - S_t^* \approx -S\rho\theta\varepsilon_t.$$

When current expectations are overly optimistic, $\theta\varepsilon_t > 0$, bonds are overpriced. In the future, bond payouts are systematically disappointing, causing low returns. The converse holds when expectations are pessimistic. In sum, DE account for (i) overoptimism in good times, (ii) excess shifts in the supply of capital and hence in spreads, which (iii) then revert together with bond returns.

Existing business cycle research has sought to produce such “excess” macroeconomic volatility under RE by introducing ingredients such as news shocks (Barsky and Sims 2011; Bianchi 2011), ambiguity (Bianchi, Ilut, and Schneider 2018), and uncertainty (Bianchi and Melosi 2016; Bloom et al. 2018; Arellano, Bai, and Kehoe 2019) in complete or incomplete market neoclassical models. Another approach emphasizes time-varying risk preferences due to habits (Campbell and Cochrane 1999), long-run risk (Bansal and Yaron 2004), or disaster risk (Barro 2006). The DE approach outlined above, and developed in Section III, has two advantages compared to these mechanisms.

First, DE can jointly match market outcomes and beliefs data showing departures from rationality, as documented in Table 2. Expectations play a key role in macroeconomics, and building models based on counterfactual expectations is problematic. Second, the empirical realism of DE yields new implications. For instance, under RE, a positive news shock produces an investment boom but not a concurrent output boom, yielding a counterfactual negative correlation between consumption and investment. Under DE, by contrast, excess optimism is produced by a current TFP boom, so current consumption can also increase. DE also produce, compared to RE models, sharp reversals of beliefs in the absence of news, which allows to separate macro volatility from the volatility of TFP, which appears limited in the data (Fernald 2014).¹¹

III. Neoclassical Model with Diagnostic Firms and Lenders

A unit mass of firms with different and persistent productivities decides whether to default, hire labor, invest, issue equity, and borrow subject to capital adjustment costs. Credit is supplied by a continuum of risk-neutral lenders. The only difference from

¹¹ Future work may explore the interaction between DE and uncertainty (Bianchi, Ilut, and Saijo 2024).

a workhorse neoclassical model with firm heterogeneity and risky debt (Khan and Thomas 2008; Arellano, Bai, and Kehoe 2019; Gilchrist, Zakrajšek, and Sim 2014) is that firms and lenders form expectations diagnostically.

Our main analysis is in partial equilibrium: We take the risk-free rate R and the wage rate W as given. The exogenous risk-free rate reflects a deliberate methodological choice. In conventional macro models the risk-free rate is endogenized by adding curvature to the utility function. We abstract from these ingredients because, as already argued, time-varying risk preferences are not externally measured. In addition, standard macro models with utility curvature produce a procyclical risk free rate, which is counterfactual (Winberry 2021; Cooper and Willis 2015; Bachmann, Caballero, and Engel 2013).¹²

The exogeneity of the wage rate is just a simplifying assumption. We relax it in Section VI, where we introduce elastic labor supply. We find that in good times overoptimistic beliefs boost labor demand and the real wage. This effect could dampen the beliefs-driven cycle, but, as we show, DE continue to create significant nonlinearity in this case as well.

A. Firms

Time is discrete. We use $'$ to denote future values and $_{-1}$ to indicate lagged values. Uppercase letters refer to macro or common values, lowercase letters to idiosyncratic objects. The generic firm has micro-level TFP z and is subject to macro level TFP A . It uses capital k and labor n as inputs to produce output according to a decreasing returns technology

$$y = Azk^{\alpha}n^{\nu}, \quad \alpha + \nu < 1.$$

Capital evolves based on investment i , which entails a one-period time to build

$$k' = i + (1 - \delta)k, \quad 0 < \delta < 1.$$

Investment entails quadratic adjustment costs $AC(i, k) = \frac{\eta_k}{2} \left(\frac{i}{k}\right)^2 k$, where $\eta_k > 0$. The log of micro TFP follows the AR(1) process:

$$(9) \quad \log z' = \rho_z \log z + \varepsilon'_z, \quad \varepsilon'_z \sim N(0, \sigma_z^2), \quad 0 < \rho_z < 1,$$

while the log of macro TFP follows

$$(10) \quad \log A' = \rho_A \log A + \varepsilon'_A, \quad \varepsilon'_A \sim N(0, \sigma_A^2), \quad 0 < \rho_A < 1.$$

¹²Furthermore, contrary to the idea that interest rate changes would smooth out micro nonlinearities (Khan and Thomas 2008), recent work suggests that nonlinearities survive in the presence of plausibly low investment elasticities to interest rates (Koby and Wolf 2020), consistent also with our structural estimates.

In each period, firms and lenders form diagnostic beliefs about a firm's productivity next period. Given the AR(1) processes (9) and (10), and given equation (2), beliefs over micro and macro TFP are described by the lognormal processes:

$$(11) \quad \log z' | (\log z, \varepsilon_z) \sim N[\rho_z(\log z + \theta \varepsilon_z), \sigma_z^2]$$

$$(12) \quad \log A' | (\log A, \varepsilon_A) \sim N[\rho_A(\log A + \theta \varepsilon_A), \sigma_A^2].$$

When $\theta > 0$, the agent forecasts next period's productivity by overweighting current news, as if the true productivity process follows an ARMA (1,1). We impose the same degree of overreaction to fluctuations in micro and macro productivity. This provides tight discipline to the model, requiring that the degree of overreaction estimated in Section IV from idiosyncratic fluctuations in firm-level profits at the same time lies in the ballpark of existing estimates of θ from macroeconomic forecasts and is capable of producing realistic aggregate boom-and-bust cycles. The single value $\theta > 0$ is thus the only difference between our model and a workhorse heterogeneous firms macrofinancial model.

We assume that, in each period, firms form expectations about next period's productivity according to equations (11)–(12) and expect to form one-step-ahead forecasts based on future productivity shocks using the very same ARMA (1,1) formula. This assumption allows us to write the firm optimization problem as a recursive and time-consistent dynamic program, conveniently simplifying computations compared to the alternative formulation in which agents optimize in each period based on a full term structure of diagnostic beliefs at different horizons.¹³

Firms act competitively. In each period, each firm first decides whether to default on its debt. If a firm defaults, its assets net of deadweight default costs are recovered by lenders, and the firm restarts with zero capital and debt after one period. If a firm repays, it also hires labor at wage W , chooses how much to invest, how much one-period debt to issue, and pays a dividend to its shareholders or issues equity. Firms maximize the expected discounted sum of current and future payouts, where the discount rate $(R)^{-1} < 1$ reflects the risk-free rate R .

Consider the firm's problem, starting from the last stage. The firm's dividend d is given by the standard formula:

$$(13) \quad d = (1 - \tau)[y - Wn - AC(i, k) - \phi] + q^\theta(s, k', b')b' - i - b + \tau((R - 1)b + \delta k),$$

which includes profits, given by the firm's output minus the wage bill, the adjustment cost, and a fixed production cost $\phi > 0$, net of the corporate income tax rate $\tau \in (0, 1)$. It also includes the resources raised by issuing new debt b' priced by the

¹³This is equivalent to assuming that agents' time t beliefs about A_{t+s} is the product $\Pi_{j=1}^s f^\theta(A_{t+j} | A_{t+j-1}, \varepsilon_{A,t+j-1})$ of the one-step conditional distributions between t and $t + s - 1$ computed along the paths of shocks that may materialize. This "factorized" distribution has the same mean as the time t diagnostic distribution about TFP at $t + s$, $f^\theta(A_{t+s} | A_t, \varepsilon_{A,t})$, but has larger variance. This is due to overreaction of one-step-ahead beliefs to possible news in the interim periods, which does not arise in the long-term forecast because the latter takes into account that news in the interim periods are zero on average.

schedule q^θ , minus the investment cost i and debt repayment b . Finally, it includes the tax rebates for capital depreciation and interest expenses on debt.¹⁴

If dividends are negative, $d < 0$, the firm issues equity. Following Gomes (2001), this entails a cost $IC(d) = I(d < 0)(\eta_f + \eta_d|d|)$, where $\eta_f > 0$ is the fixed and $\eta_d > 0$ is the variable cost of issuance.

The firm makes its other decisions considering four state variables: Its current micro TFP z , macro TFP A , the micro shock ε_z , and the macro shock ε_A . The exogenous news states are relevant for forming diagnostic expectations, and we collect all of the firm's exogenous states in the vector $s = (z, \varepsilon_z, A, \varepsilon_A)$. A firm is also identified by two endogenous states, its inherited capital stock k and debt b . Given an overall state (s, k, b) , the firm's problem can be written recursively. Upon entering the current period, the value of the firm is given by

$$(14) \quad V^\theta(s, k, b) = \max[V_D^\theta(s), V_{ND}^\theta(s, k, b)],$$

where $V_{ND}^\theta(s, k, b)$ is the continuation value from not defaulting and $V_D^\theta(s)$ the continuation from defaulting. Condition $V_{ND}^\theta(s, k, b) < V_D^\theta(s)$ identifies states in which the firm optimally defaults. The continuation value from not defaulting is recursively determined as

$$(15) \quad V_{ND}^\theta(s, k, b) = \max_{k', b', n} \left\{ d - IC(d) + \frac{1}{R} \mathbb{E}^\theta[V^\theta(s', k', b') | s] \right\}.$$

If the firm does not default, it hires labor n , capital k' , and sets debt b' to maximize its current dividend plus its diagnostically expected discounted future value $V^\theta(s', k', b')$. The labor choice n is statically optimized, leaving only the intertemporal choices of k' and b' .

If the firm defaults, its assets k net of deadweight costs are claimed by lenders during a period of reorganization with no production. The firm restarts with zero debt and assets:

$$(16) \quad V_D^\theta(s) = \left\{ 0 + \frac{1}{R} \mathbb{E}^\theta[V(s', 0, 0) | s] \right\}.$$

After defaulting, a firm must borrow to invest. Equations (14), (15), and (16) determine the optimal default policy by $df^\theta(s, k, b)$ and, for nondefaulting firms having $df^\theta(s, k, b) = 0$, the policies for endogenous states $k'^\theta(s, k, b)$ and $b'^\theta(s, k, b)$.

B. Lenders

Firms borrow from risk-neutral, deep-pocketed lenders whose required expected return is the risk-free rate R . If a firm (s, k, b) defaults on debt b , the lender receives the recovery rate:

$$\mathcal{R}(k, b) = (1 - \tau)\gamma \frac{(1 - \delta)k}{b},$$

¹⁴For computational simplicity, we assume the rebate is on average equal to the risk-free rate $R - 1$.

which reflects, net of tax, an exogenous fraction γ of the liquidation value $(1 - \delta)k$ of the firm's capital stock. The remaining fraction $1 - \gamma$ is a deadweight loss.

The price of debt $q^\theta(s, k', b')$ adjusts endogenously so that the diagnostically expected bond return is equal to the risk-free rate R :

$$(17) \quad q^\theta(s, k', b') = \frac{1}{R} \mathbb{E}^\theta \left[1 + df^\theta(s', k', b') (\mathcal{R}(k', b') - 1) \mid s \right].$$

To equalize expected bond returns across firms, riskier firms promise a higher interest rate.¹⁵ Thus, the firm's interest rate spread relative to the risk-free rate is given by

$$S^\theta(s, k', b') = \frac{1}{q^\theta(s, k', b')} - R.$$

These equations illustrate how diagnosticity affects spreads. On the demand side, diagnosticity affects the firm's default $df^\theta(s, k, b)$, debt $b^\theta(s, k, b)$, and investment $k^\theta(s, k, b)$ policies. On the supply side, diagnosticity affects the probability of default perceived by lenders, as captured by the operator $\mathbb{E}^\theta(\cdot)$ in (17). The interaction between demand and supply plays a key role in producing macroeconomic fluctuations.

C. Solving the Model

A partial equilibrium is a collection of (i) firm policies b^θ , k^θ , and df^θ ; (ii) firm values V_{ND}^θ , V_D^θ , and V^θ ; and (iii) a lender debt price schedule q^θ such that (iv) taking as given debt prices q^θ , firm policies and values satisfy equations (14), (15), and (16); (v) taking as given firm policies, debt prices satisfy the zero-profit condition in equation (17); and (vi) expectations in the firm and lender problems $\mathbb{E}^\theta$ reflect the dynamics in equations (11) and (12).

We solve the model numerically. In addition to the set of Bellman equations for V^θ , V_{ND}^θ , and V_D^θ , the model features an equilibrium fixed point between firm default policies df^θ and credit prices q^θ in equation (17). We employ an iterative approach detailed in Supplemental Appendix A. First, we guess a firm default rule df^θ , computing the implied debt price schedule q^θ according to the lenders' zero-profit condition. Then, we solve the Bellman equations for V^θ , V_D^θ , and V_{ND}^θ using discretization and policy iteration. If the implied default states, that is, those with $V_{ND}^\theta < V_D^\theta$, match the set of initial guesses, then the iteration is complete. Otherwise, we compute the newly implied default states and repeat the process. The algorithm we employ is standard within the literature solving quantitative dynamic corporate finance models and follows the implementation in Strebulaev and Whited (2012).¹⁶

¹⁵The realized firm bond return is given by $R^{firm}(s, s', k', b') = \frac{1 + df^\theta(s', k', b') (\mathcal{R}(k', b') - 1)}{q^\theta(s, k', b')}$.

¹⁶Our numerical approach here is highly computationally intensive, given the presence of four exogenous states, two endogenous states, an endogenous default rule, and a debt-pricing fixed point. However, judicious application of parallelization and an economical approach to storage of micro-level outcomes following Young (2010) and Terry (2017) allow for solution of the model in several minutes on a desktop computer.

TABLE 3—EXTERNALLY FIXED PARAMETERS

	Parameter	Role	Value	Source
1	δ	Depreciation rate	0.10	Annual solution
2	R	Risk-free rate	1.04	Annual solution
3	α	Capital elasticity	0.25	Bloom et al. (2018)
4	ν	Labor elasticity	0.50	Bloom et al. (2018)
5	ρ_A	Macro TFP autocorrelation	0.95	Bloom et al. (2018)
6	τ	Corporate income tax	0.20	Effective rates, CBO (2017)

Note: The table reports the parameter symbol, numerical value, a description, and source information for each of the externally fixed parameters.

DE create a wedge between the value functions of two firms reaching the same current productivity via different news: The firm experiencing good recent news displays overoptimism and hence an inflated continuation value relative to a firm experiencing bad recent news (see Supplemental Appendix Figure A.1). This effect on the demand side for credit, combined with diagnostic shifts in the supply of capital by lenders, leads to aggregate effects.

IV. Estimating the Model and Inspecting the Mechanism

A. Estimation

Our model has 16 parameters, listed in Table 3 and Table 4. The six parameters in Table 3 are set to conventional values for a model like ours. The remaining 10 parameters: ρ_z and σ_z govern the micro-level TFP process; η_k and ϕ capture the adjustment and operating costs, γ the lenders' recovery rate; σ_A encodes macro volatility, η_f and η_d the equity issuance costs. θ is the key diagnosticity parameter. We also allow for and estimate the volatility σ_π of i.i.d. measurement noise in profits, which captures spurious variability in measured profits due to accounting conventions.

Using the simulated method of moments (SMM), we structurally estimate these 10 parameters using 16 statistics. Three of them are macro moments: the average credit spread, the average frequency of default, and GDP growth volatility. Macro data on default and spreads encode information about the fixed cost ϕ and recovery rate γ ; GDP growth volatility is informative about macro shocks σ_A . The remaining 13 moments are at the firm level. One set of firm-level moments comes from the variance-covariance matrix of profits, investment rates, spreads, and debt issuance. Profits and their correlations yield information about firm-level productivity and noise, helping to identify σ_z , ρ_z , and σ_π . Firm investment also reflects adjustment costs, helping to identify η_k . Debt issuance, together with credit spreads, aid in the identification of equity issuance costs η_f and η_d .

Three additional moments use data on forecast errors, whose use is the key innovation in our estimation, directly disciplining the degree of overreaction θ . One moment is the standard deviation of forecast errors. The other two moments are the correlation of the change in future forecast errors with current changes in the investment rate and debt issuance. Moments based on within-firm changes allow for systematic heterogeneity across firms, that is, firm fixed effects, while still naturally linking to overreaction. Suppose that investment i_t , debt issuance b_t' , and profits π_t

TABLE 4—PARAMETER ESTIMATES

Parameter	Role	DE model	RE model
1 θ	Diagnosticity	0.913 (0.097)	—
2 ρ_z	Micro TFP autocorrelation	0.670 (0.024)	0.924 (0.011)
3 σ_z	Micro TFP volatility	0.251 (0.010)	0.096 (0.004)
4 η_k	Capital adjustment cost	4.023 (0.329)	3.490 (0.174)
5 ϕ	Fixed operating cost	0.155 (0.014)	0.075 (0.002)
6 γ	Recovery rate	0.233 (0.102)	0.177 (0.275)
7 σ_A	Macro TFP volatility	0.006 (0.001)	0.008 (0.002)
8 η_f	Equity fixed cost	0.012 (0.004)	0.076 (0.003)
9 η_d	Equity linear cost	0.109 (0.028)	0.100 (0.223)
10 σ_π	Earnings noise	0.678 (0.172)	0.390 (0.077)

Notes: The table reports point estimates and standard errors for each of the parameters in our SMM estimation of both the DE model and the RE model. The moment covariance matrix is based on firm-level clustering in the micro block and a stationary block bootstrap in the macro block. The moment Jacobian is computed numerically. In the SMM estimation, the weighting matrix is optimal, that is, the inverse of the moment covariance matrix.

are linearly increasing in expected log TFP, with positive slope coefficients respectively given by a_i , a_b , and a_π . Using equation (3), the covariance between the future change in forecast errors and the current change in investment and debt is then given by

$$(18) \quad \text{cov}[(\pi_{t+2} - E_{t+1}^\theta \pi_{t+2}) - (\pi_{t+1} - E_t^\theta \pi_{t+1}), x_t - x_{t-1}] = a_\pi a_x \rho \theta (1 + \theta),$$

$$x = i, b.$$

Equation (18) is positive if and only if expectations are diagnostic, $\theta > 0$. If beliefs overreact, good news leads to systematic disappointment next period, but no disappointment two periods from now, causing forecast errors to grow between $t + 1$ and $t + 2$.¹⁷ The RE model is instead unable—because of its more general failure to generate forecast error predictability—to generate any comovement between future forecast error growth and investment or debt changes of the firm.

Table 5 reports the moments we target in the Data column. We obtain these moments using our combined datasets with firm financials, earnings forecasts, and credit spreads described in Section I. Just as in Section I, we focus on idiosyncratic variation, residualizing the underlying investment, debt, and spread series with respect to firm and time effects. As seen in Section I, the micro data show overreaction, as reflected in the positive forecast error growth covariances. To estimate the model parameters, we minimize the deviation of the empirical moments in Table 5 from those computed in a comparable unconditional simulation of the model. We weight the moments optimally using the inverse of our estimate of the moment covariance matrix, implying an asymptotically efficient SMM estimator. See Supplemental Appendix B for a more detailed description of the variable

¹⁷ Supplemental Appendix B offers a detailed analysis of this property and further derivations of moment conditions in closed form in the linear case discussed above. Note that while our quantitative model is clearly nonlinear, and the formula in equation (18) is therefore approximate, the intuition still holds in the more general model and proves useful for identification of the parameter θ . In particular, the covariance moments grow with the value of θ in our DE simulations.

TABLE 5—ESTIMATED MODEL FIT

Moment		Data	DE model	RE model
<i>Panel A. Micro moments</i>				
1	cov (Δ Forecast error $_{t+1}$, Δ Investment $_{t-1}$)	0.003	0.005	0.000
2	cov (Δ Forecast error $_{t+1}$, Δ Debt $_{t-1}$)	0.004	0.010	0.000
3	std. dev (Forecast error $_{t+1}$)	0.305	0.207	0.200
4	std. dev (Profit $_t$)	0.262	0.181	0.197
5	corr (Profit $_t$, Investment $_t$)	0.257	0.423	0.125
6	corr (Profit $_t$, Debt $_t$)	0.120	0.479	0.111
7	corr (Profit $_t$, Spread $_t$)	−0.159	−0.039	0.025
8	std. dev (Investment $_t$)	0.067	0.066	0.040
9	corr (Investment $_t$, Debt $_t$)	0.104	0.807	0.515
10	corr (Investment $_t$, Spread $_t$)	−0.057	−0.071	0.135
11	std. dev (Debt $_t$)	0.112	0.109	0.101
12	corr (Debt $_t$, Spread $_t$)	−0.036	−0.025	0.204
13	std. dev (Spread $_t$)	0.011	0.007	0.006
<i>Panel B. Macro moments</i>				
14	$\mathbb{E}$ (Spread $_t$)	0.029	0.023	0.004
15	$\mathbb{E}$ (Default $_t$)	0.003	0.005	0.002
16	std. dev (Δ GDP $_t$)	0.015	0.013	0.017

Notes: The moments were computed on a sample combining information from the Compustat, IBES Manager Guidance, and FISD/Trace Bond databases. For the micro moments, the forecast error, profit, and investment, series are expressed relative to firm tangible capital stocks, while the debt series is scaled by total assets and the spread is in proportional units. For the macro moments, the mean spread is the average across years of the mean spread across firms in the FISD/Trace Bond-Compustat merged database; mean default is the average across years of the mean default rate across firms in the FISD/Trace-Compustat merged database. The GDP series is annual GDP in chained 2012 dollars.

definitions, sample construction, and our approach to computing the SMM point estimates and standard errors. Given our use of 10 parameters to target 16 moments, this is a highly overidentified structural estimation of a nonlinear model. We can exploit a great deal of information but are not in general able to deliver an exact fit.

Table 4 reports the SMM point estimates and standard errors for our DE model. The diagnosticity parameter $\theta \approx 0.9$ is in the ballpark of the values found by Bordalo, Gennaioli, and Shleifer (2018) using data on professional forecasts of credit spreads ($\theta = 0.9$); by Bordalo et al. (2019) using analyst expectations of US listed firms' long-term earnings growth ($\theta = 0.9$); by Pflueger, Siriwardane, and Sunderam (2020) using stock price-derived measures of risk perception ($\theta = 1$); and by Bordalo et al. (2020) using professional forecasts of several macro series ($\theta = 0.5$). Intuitively, a value of θ near 1 means that forecast errors are roughly equal to the size of incoming news.

The estimated values governing physical factors such as micro TFP volatility σ_z and capital adjustment costs η_k are close to those from other work estimating firm-level shock processes with similar data (Gourio and Rudanko 2014; Terry 2023; Khan and Thomas 2008). The parameters governing financial frictions indicate equity issuance costs η_f , η_d and recovery rates γ comparable to those in Hennessy and Whited (2007). The fixed operating costs ϕ in model units is linked to average default rates and spreads. Finally, the estimated measurement error in profits σ_π suggests that a high degree of noise in accounting conventions is needed to match the covariance of profits and other firm-level outcomes.

Using the same micro and macro moments, we also conduct a constrained SMM estimation exercise to pin down the nine parameters of an RE model in which we set $\theta = 0$, eliminating diagnosticity in beliefs. We do not expect this model to succeed, for it abstracts from many dimensions, such as financial intermediaries, etc. However, it allows us to quantify the improvement obtained when adding only the overreaction parameter θ . Point estimates and standard errors for the RE model are reported in Table 4. In the DE model we estimate a substantially lower persistence ρ_z of micro TFP shocks and a lower aggregate TFP volatility σ_A compared to the RE model. After good news, the DE manager becomes overoptimistic about the future, which boosts investment “as if” the current shocks were more persistent. DE is also a source of “shocks,” which reduces the estimated σ_A : When optimism wanes, investment is cut “as if” a negative shock hit. In addition, DE also inflates perceived volatility. Because production function–based payoffs are convex, this implies that a firm’s average incentives to default are also lower. To match the observed default frequency, the DE model estimates a higher fixed operating cost ϕ . Overall, the fact that the estimates of both θ and several other parameters match those in previous work and the fact that DE reduces the required volatility of TFP shocks suggest that DE may provide a useful and tractable account of the data.

We next turn to moment fit, reported in Table 5. The DE model fits the micro moments well, especially considering its strong nonlinearity and the overidentified SMM estimation. Critically, DE can obtain predictable forecast errors, while the RE model cannot. This will prove critical for producing excess macroeconomic instability. Realistically high belief volatility also allows the DE model to fit higher investment volatility and, importantly, to match the negative correlation of credit spreads with profits, investment, and debt. In the RE model, inability to match beliefs yields counterfactually procyclical spreads. The intuition is simple: With DE, lenders’ overoptimism after good news causes a stronger outward credit supply shift, lowering spreads even in the face of increased debt or credit demand, which does not occur under RE.

One discrepancy of the DE model with the data is that it produces a counterfactually high correlation between debt and both investment and profits. We speculate that an extended version of our model allowing for empirically realistic costs of debt financing, such as restrictive contractual covenants, would moderate debt movements and further improve the DE model’s fit. Overall, though, Table 5 shows the ability of the DE model to match realistic belief overreaction and therefore moments tied to macrofinancial volatility.

We now show that these differences between the DE and RE models prove critical to account for boom-and-bust cycles, both at the aggregate and firm levels.

B. Model Mechanisms

Nonlinearity in the Investment Response to TFP Shocks.—To study the real implications of DE, we simulate the contemporaneous response of macro investment to a negative macro TFP shock. We compute this response for different initial conditions, captured by TFP shocks of varying magnitudes in the previous period. This exercise highlights one important consequence of DE: fragility after good times. This nonlinearity is critical in our framework to account for boom-and-bust credit cycles.

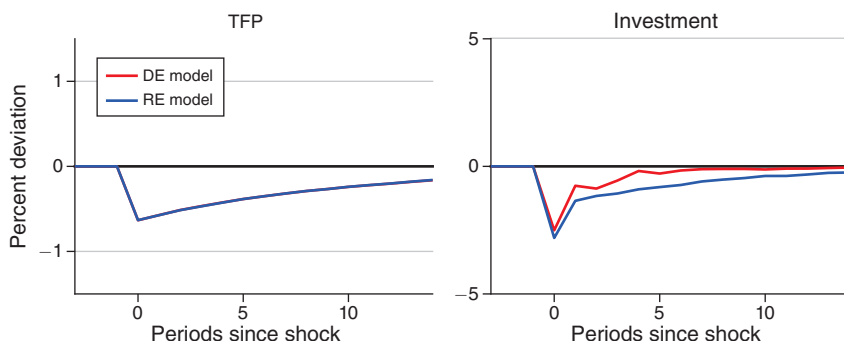


FIGURE 1. INVESTMENT RESPONSE TO A NEGATIVE TFP SHOCK

Notes: The figure plots the simulated generalized impulse responses of investment (right panel) to a 1 standard deviation negative shock to macro TFP (left panel) occurring at period 0. The DE model (red) and RE model (blue) paths are based on identical shocks to TFP over 10,000 simulated experiments following the Koop, Pesaran, and Potter (1996) methodology.

Figure 1 plots the average impulse response of investment to a negative shock to macro TFP in the DE model (red line) and RE model (blue line).¹⁸ These responses reflect “typical times,” based on simulated reactions to negative TFP shocks across 10,000 experiments following the methodology of Koop, Pesaran, and Potter (1996). In normal times, the investment responses to TFP shocks differ only slightly across the DE and RE models.

We next simulate the impact of the same negative shock to macro TFP varying the initial conditions. Figure 2 reports the investment response in the DE model (vertical axis, red line) as a function of the magnitude of the previous period’s macro TFP shock (horizontal axis). There is a strong nonlinearity taking the form of “fragility in good times”: The same adverse TFP shock is much more damaging for aggregate investment when it occurs in good times. Figure 2 adds to the same diagram the response of investment in the RE model (blue line). The latter is almost the same across initial conditions, with little nonlinearity or fragility in good times. Recent work shows that investment is more sensitive to shocks during booms than during normal times (Bachmann, Caballero, and Engel 2013; Winberry 2021; Bloom et al. 2018). Figure 2 shows that belief overreaction can produce this feature. Here, nonlinearity arises because even a small negative shock can induce a sharp reversal after good times. DE models with longer overreaction lags, $J > 1$, can further amplify reversals, as shown by Bianchi, Ilut, and Saijo (2023).

The investment response in Figure 2 combines two forces: the demand for capital reflecting managers’ expectations of future profits and the supply of capital reflecting lenders’ expectations of default. In good times, diagnostic managers are overly optimistic, so they borrow and invest more than under RE. Diagnostic lenders are also overly optimistic and hence are more willing to lend than under RE. Under DE, then, both the demand and the supply of credit overreact in good times, which causes a leverage and investment boom. When a negative shock hits, managers and

¹⁸The shock is identically sized in the two experiments as a 1 standard deviation shock to macro TFP using the parameter estimate $\hat{\sigma}_A$ from the DE model.

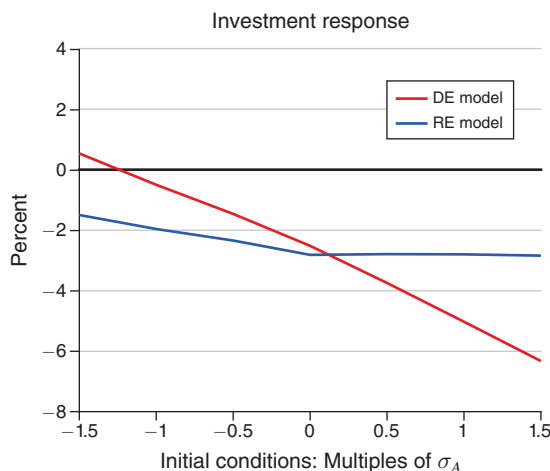


FIGURE 2. INVESTMENT NONLINEARITY

Notes: The vertical axis in the figure reports the simulated first-period impulse response of macro investment to a 1 standard deviation negative shock to macro TFP, and the horizontal axis reports the initial conditions, that is, the magnitude of the shock to macro TFP in the previous period. Both the DE model (red line) and RE model (blue line) are reported on the figure.

lenders become too pessimistic, both the demand and the supply of credit contract, and firms drastically cut investment.¹⁹ The effect is drastic because, starting from a position of overoptimism, firms are overindebted to begin with.²⁰

One question is whether credit supply or demand play a larger role in creating fragility after good times. To shed light on this issue, we impose that lenders are fully rational—they have $\theta = 0$ —while θ is unconstrained for firms. Reestimating the model for this case, we simulate a set of “rational pricing” responses of investment to a negative TFP shock comparable to those in Figure 2. Qualitatively, the response of investment displays the nonlinearity typical of the DE model, with a quantitatively dampened magnitude in Supplemental Appendix Figure A.2.²¹ We conclude that under DE, shifts in the supply of credit play an important role in creating fragility during booms, especially due to overreaction to aggregate shocks, a pattern that is also evident in the next exercise.

Financial Reversals.—A second related implication is that DE can produce strong macrofinancial reversals without large negative TFP shocks. We simulate the model unconditionally for 50,000 periods and consider large financial reversals, defined as

¹⁹Interestingly, state dependence in Figure 2 also generates a positive response during bad times under DE, when even a moderate negative shock to TFP can be considered “good news” relative to overpessimistic beliefs. Under RE, the response curve is uniformly negative.

²⁰One can decompose the DE nonlinearity in Figure 2 into contributions from direct belief reversals versus endogenous increases in the riskiness of overoptimistic firms’ states k and b . Supplemental Appendix Figure A.4 reveals that both channels play a meaningful quantitative role.

²¹In Supplemental Appendix Figure A.3, we reproduce investment responses for another special case, one with lender DE over aggregate shocks and RE over firm-level shocks. The resulting aggregate investment nonlinearity is similar to our baseline DE case.

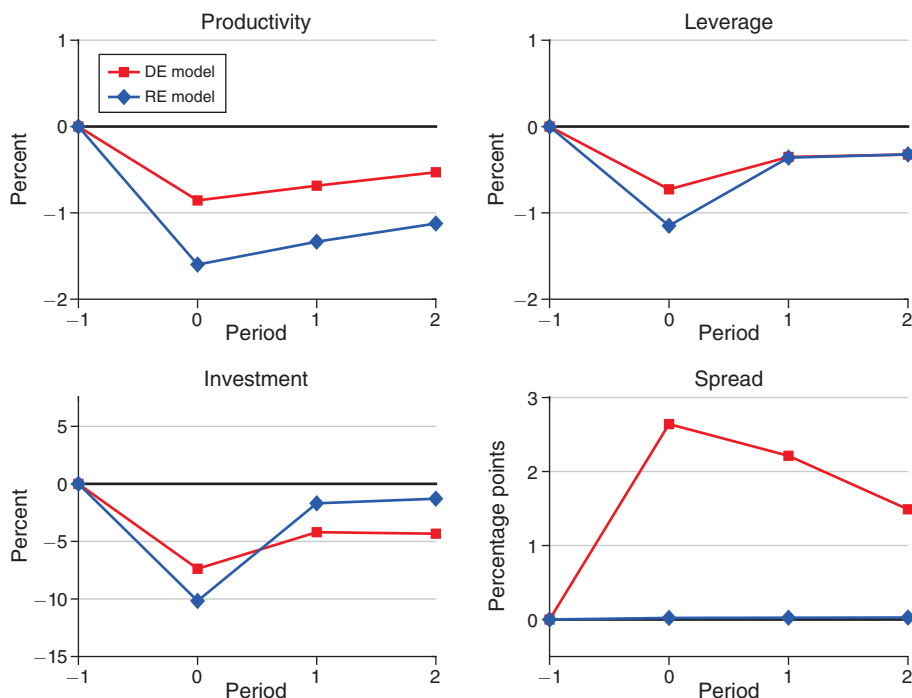


FIGURE 3. FINANCIAL DISRUPTIONS ON AVERAGE

Notes: The figure plots the average path of macro TFP A (top left), macro leverage B'/K (top right), macro investment I (bottom left), and the macro spread $\mathbb{E} \text{ Spread}$ (bottom right) around periods of financial disruption from an unconditional simulation of the DE model (red line) versus the RE model (blue line). The variables are plotted in percent (TFP, credit, and investment) or percentage point (spread) deviations from the pre-disruption period. A financial disruption, dated at period t on the horizontal axis, is a period with spread growth in the top 10 percent and investment growth in the bottom 10 percent of realizations in the simulated data for each model.

periods in which spread growth is in the top 10 percent of the distribution and investment growth is in the bottom 10 percent. We extract the average dynamics of TFP, leverage, investment, and spreads around these episodes (labeled as period “0”) for both the DE and the RE model and report them in Figure 3.

Consistent with existing evidence, in the DE model (red lines), reversals arise after only a mild decline in TFP (top left panel) relative to the RE case (blue lines). Despite a relatively small fundamental TFP disruption, the DE model generates declines in leverage (top right panel) and investment (bottom left panel) similar to the RE model.²² And, unlike the RE model, which generates hardly any movement in credit spreads during financial disruptions (bottom right panel), the DE model exhibits large increases. Two forces are at work. First, overreaction leads pessimistic firms under DE to cut back severely on investment and financing even in the face of only small changes in TFP. Second, under DE but not under RE, credit supply also

²²Note that in the particular exercise in Figure 3, we find that large reversals can be generated by small negative TFP shocks, but the broader DE framework can in fact generate even richer dynamics. For example, in Bianchi, Ilut, and Saijo (2023), fluctuations are sometimes driven by disappointment of overoptimistic expectations alone, rather than through the direct negative impact of TFP shocks, a fact that demonstrates another potential advantage of DE versus RE models.

TABLE 6—MACRO CREDIT SPREAD FLUCTUATIONS

	Standard deviation (1)	Corr. with output (2)
Data	0.011	−0.237
DE model	0.014	−0.563
RE model	0.001	−0.627

Notes: The table reports the standard deviation (column 1) and the correlation with aggregate output (column 2) of the aggregate credit spread. Empirically, output is real GDP from the US NIPA accounts, and the credit spread is the mean across firms in a given year in the merged Compustat-IBES-FISD/TRACE sample. Empirically, output is HP-filtered in logs with smoothing parameter 100. The model moments are computed from an unconditional simulation for the DE model (middle row) and the RE model (bottom row), with an identical set of aggregate TFP shock draws for both model versions. Model quantities refer to the total value (output) or average value (spread) computed from the distribution of firms. Units are in proportional terms; that is, 0.01 = 1%.

shifts downward in an overly pessimistic fashion, boosting spreads even in the face of a decline in aggregate leverage. This confirms the importance of overreacting lenders to account for shifts in the supply of capital and hence in credit spreads.

V. Quantitative Implications

We examine the model's ability to quantitatively match two sets of untargeted facts. The first set concerns our motivating evidence on (i) macro credit spread fluctuations and (ii) the boom-and-bust dynamics of beliefs, returns, and investment documented in Section I. The second set of facts concerns movements of credit spreads and other aggregates around the financial crisis of 2008.

A. Volatile Credit Spreads

Consider the ability of our model to reproduce macro credit spread fluctuations, which are entirely untargeted in our calibration procedure. Table 6 reports both the volatility of the mean credit spread (column 1) and its cyclical correlation with output (column 2) in the data, the DE model, and the RE model. The DE and RE model moments are computed based on an identical set of exogenous shocks; that is, we feed the same sample paths for TFP into the two models. In the data, the standard deviation of credit spreads is about 1 percent annually. Unlike the RE model, which is unable to produce meaningful volatility (generating a standard deviation of only 0.1 percent), the DE model generates substantial fluctuations in line with the data, with a standard deviation of around 1.4 percent. The DE model's aggregate credit spread fluctuations are realistically countercyclical (column 2). The RE model also produces aggregate spread countercyclical but with too little volatility to be quantitatively meaningful.

This result shows the promise of DE for producing realistic macrofinancial volatility (and in particular countercyclical spreads) in business cycle models. It is thus instructive to consider the contribution of a “DE block” using the language of “wedges,” in the spirit of Chari, Kehoe, and McGrattan (2007). Supplemental

Appendix Section A.7 analytically shows in a simplified version of our model that DE generate countercyclical fluctuations in the investment wedge, with a standard deviation of around 1 percent per year.²³ That is, overoptimism during good times causes a DE firm to act as if its investment price was low from the perspective of a business cycle accountant imposing belief rationality, and the reverse during good times.

The mapping of DE to an investment rather than, say, an efficiency or labor wedge is important. Investment wedges are often the target of macrofinancial models (Kiyotaki and Moore 1997; Justiniano, Primiceri, and Tambalotti 2011; Brinca et al. 2016), and DE can produce realistic fluctuations in them and hence in macrofinancial outcomes thanks to its ability to produce realistic movement in beliefs. This makes our model particularly useful for understanding conditional facts such as large, predictable macrofinancial reversals after good times.²⁴

Of course, DE alone cannot generate a perfect fit of untargeted business cycle moments, which also reflect factors (aggregate demand, price rigidities, etc.) from which we abstract here. In particular, mechanisms moving labor and efficiency wedges such as uncertainty shocks and nominal rigidities (Christiano, Motto, and Rostagno 2014) are also needed. An important issue is whether a combination of these forces with DE can improve our understanding of unconditional business cycle variation, as some initial work seems to show (Bianchi, Ilut, and Saijo 2023; L'Huillier, Singh, and Yoo 2024).

B. Boom-and-Bust Cycles in Financial and Real Activity

We next assess the explanatory power of the model for the firm-level boom-and-bust cycles documented in Section I as well as its ability to account for aggregate (i.e., sector-level) cycles. Consider firm-level cycles first. We simulate the DE and RE models for many periods and then use the simulated data to reestimate the two-stage regressions of Table 2 in Section I. In the first stage, we regress future profit forecast errors on current investment. In the second stage, we use these fitted errors as an explanatory variable for future bond returns on the financial side and investment growth on the real side. Table 7 reports the results. In line with the estimates of Table 2, in the DE model high investment predicts future negative forecast errors in column 1, indicating that expectations are overly optimistic in good times. Of course, the RE model (unreported) fails completely, by construction, to generate forecast error predictability of the type seen in column 1.

In columns 2 and 3, the DE model also captures the positive correlation between predicted forecast errors, bond returns, and investment growth: When borrowers and lenders are overly optimistic (their profit forecasts are systematically above reality), realized bond returns and investment growth are lower. Quantitatively, comparing Table 7 from simulated data to the estimates from Table 2, the DE model

²³ Unsurprisingly, in our model with frictionless static labor demand, DE does not cause fluctuations in labor or efficiency wedges in the analysis in Supplemental Appendix Section A.7.

²⁴ These conditional implications of DE are linked to movements in underlying investment wedges. Our measure of the investment wedge derived in Supplemental Appendix Section A.7 moves by about 3 standard deviations over the full range of Figure 2, by about one-and-a-half standard deviations during the financial disruption in Figure 3, and by about 2 standard deviations in the Great Recession exercise we conduct in Figure 4.

TABLE 7—LINKING FORECAST ERRORS AND MICRO REVERSALS IN THE MODEL

Dependent variable:	Fcst. error _{<i>t</i>+1}	Return _{<i>t</i>+1}	Δ Investment _{<i>t</i>+1}
IV stage:	First (1)	Second (2)	Second (3)
Forecast error _{<i>t</i>+1}		0.011 (0.001)	1.108 (0.004)
Investment _{<i>t</i>}	−0.860 (0.007)		
Year effects	✓	✓	✓
Profit control			✓
Firms	1,000	1,000	1,000
Firm-years	250,000	250,000	250,000
First-stage <i>F</i>	15,146		

Notes: The table reports first- and second-stage estimates based on simulated firm-level data from the DE model. Column 1 reports the first stage, and columns 2–3 report second-stage regressions. Standard errors are clustered at the firm level. Forecast error is realized minus expected profits normalized by the firm's capital stock. Investment is the investment rate, that is, capital expenditures normalized by the firm's capital stock. The bond return is the realized bond return. Column 3 controls for current profits. For all series, $0.01 = 1\%$.

somewhat reasonably reproduces the untargeted size of the first- and second-stage coefficients.²⁵ We conclude that firm-level reversals in real and financial conditions after good times are plausibly sized in the DE model, while being completely absent in the RE model.

Firm-level reversals need not translate into aggregate reversals. First, firm managers may be particularly attentive to firm-level news, not aggregate news, and hence overreact only to the former. Second, and relatedly, the effect of overoptimism may be amplified during extreme conditions such as a firm's financial distress, but the aggregate-level consequences may be muted due to the paucity of defaulting firms. Investigating these patterns at a more aggregated level therefore offers an important test.

Studying belief dynamics at the macroeconomic level is difficult due to the relatively short post-2000 time window for our annual manager expectations dataset. As an alternative, we implement our two-stage empirical approach at the 3-digit sector level, which offers meaningful aggregation while also allowing us to exploit cross-sectoral variation. Panel A of Table 8 reports the results. In column 1, our first stage reveals that periods of high average investment in a sector predict lower average forecast errors in the subsequent year; that is, good times beget excess optimism. In columns 2–3, we see that periods of excess optimism are followed by lower bond returns and lower investment growth in the same sector. An industry in which overoptimistic firms are *predictably* disappointed by 1 standard deviation more on average, with a decline in the forecast error of 0.32, sees average bond returns decline by $0.32 \times 0.047 \approx 1.5$ percentage points (column 2) and average

²⁵ While the first-stage coefficient in Table 7's column 1 is formally untargeted, our SMM estimation procedure does target a set of closely linked firm-level covariances. In light of this, a comparison of the also untargeted second-stage coefficients in columns 2–3 to their empirical counterparts in Table 2 provides a tighter test.

TABLE 8—LINKING FORECAST ERRORS AND AGGREGATE REVERSALS

Dependent variable:	Forecast error _{t+1}	Return _{t+1}	Δ Investment _{t+1}
IV stage:	First (1)	Second (2)	Second (3)
<i>Panel A. Industry data</i>			
Forecast error _{t+1}		0.047 (0.014)	1.221 (0.050)
Investment _t	−0.476 (0.153)		
Industry effects	✓	✓	✓
Controls		✓	✓
Industries	111	111	111
Industry-years	1,291	1,291	1,291
First-stage <i>F</i>	10		
<i>Panel B. DE model</i>			
Forecast error _{t+1}		0.053 (0.002)	2.039 (0.012)
Investment _t	−0.513 (0.004)		
Industry effects	✓	✓	✓
Controls		✓	✓
Years	50,000	50,000	50,000
First-stage <i>F</i>	17,279		

Notes: The table reports first- and second-stage estimates based on industry data from the Compustat-IBES-FISD/TRACE sample at the SIC3 × fiscal year level (panel A) and simulated aggregate data from the estimated DE model (panel B). Forecast error is average realized minus expected profits normalized by capital stocks. Investment is the average investment rate, that is, capital expenditures normalized by the capital stock. Return is the average realized bond return, and spread is the average realized bond spread relative to the risk-free rate. Column 1 reports the first stage, and columns 2–3 report second-stage regressions. Standard errors are clustered at the industry level. For the purposes of rough comparison with industry data in panel A, simulated aggregate data in panel B is grouped into sequential blocks or industries with the same average length as in the data. So in panel B “Industry effects” refers to fixed effects for these groupings. Columns 2–3 control for the lagged spread and current profits relative to the capital stock. For all series, 0.01 = 1%.

investment rate declines of $0.32 \times 1.221 \approx 39$ percentage points (column 3).²⁶ There are significant boom-and-bust sectoral cycles in the data.

Even though our model lacks an explicit industry structure, we assess its ability to produce sector-level cycles by grouping long-time series of simulated aggregate data from our model into nonoverlapping sequential blocks or “industries.” We then run panel two-stage regressions, carefully maintaining symmetry between our simulated data and empirical work by (i) including industry effects and (ii) only considering blocks or industries with average length at just above 10 years, as in the data. Panel B reports the estimates based on simulated data from our estimated DE model since the unreported RE model is unable, by construction, to produce forecast error predictability. We see that the DE model reproduces the

²⁶The empirical results in panel A of Table 8 are quite robust. The exact level of aggregation turns out to not be crucial for our results. Supplemental Appendix Table B.7 reports similar results based on industry data aggregated to the two-digit level. In columns 2–3 we control for lagged spreads and profits, to maintain comparability with our firm-level analysis, but this is not essential, as shown in Supplemental Appendix Table B.8. Including industry effects is also not essential, as shown in Supplemental Appendix Table B.9.

aggregate link between good times (in which investment is high) and negative future forecast errors, reversals in realized bond returns, and reversals in investment growth. Quantitatively, the DE model decently matches the estimated coefficients in panel A, which is remarkable given the entirely untargeted nature of these aggregate boom-and-bust patterns.

We conclude that the DE model offers a good account of systematic or aggregated sources of boom-and-bust cycles in real and financial activity at the firm and aggregate levels. Targeting beliefs data improves quantitative macro models' match to the data in a parsimonious way.

C. The Financial Crisis of 2007–2009

Lastly, we study the ability of the DE model to generate the increase in credit spreads associated with the Lehman crisis in September 2008. Our model misses some important elements of large crises, such as gradual and persistent asset price inflation and runs on financial intermediaries. Asset price bubbles can be obtained under DE by introducing information frictions as in Bordalo, Gennaioli et al. (2021), but here we abstract from them to isolate the role of overreaction in the sharpest way. For the same reason, we abstract from financial intermediaries considered in Maxted (2023) and Krishnamurthy and Li (2025).

What type of TFP shock does the DE model need to exactly reproduce the observed jump in spreads in the late 2000s? What are the macro consequences of such shocks? To answer these questions, we perform a crisis decomposition exercise. We separate the 2004–2009 period into a “precrisis” period, 2004–2007, and a “crisis” period, 2008–2009. As mentioned before, our model yields neither a gradual path of asset price inflation nor the gradual spread reduction during 2004–2007, making these broader periods the more natural quantitative target for us. During these years, US credit spreads jumped from an average of 1.6 percent to 4.5 percent.

We use the simulated DE model to reverse engineer the TFP path needed to reproduce this path for credit spreads. Figure 4 reports this TFP path along with its implications for the growth of investment, corporate profit forecasts, and credit spreads. We report results for the DE and the RE model under the same TFP path. The red (blue) bars show the difference between crisis and precrisis outcomes produced by the DE (RE) model, while the green lines are data. In the bottom left panel, the DE model by construction perfectly matches the precrisis and crisis spreads and hence, by construction, the plotted spread growth. Remarkably, though, the bottom right panel shows that the DE model explains the credit spreads increase with a path for TFP growth that is very similar to measured TFP growth. This fact is noteworthy because the TFP dynamics are untargeted. With the same TFP path, the RE model does not produce any increase in the spread. Clearly, the limited ability of the RE model to provide overall spread volatility seen in Table 6 also prevents the model from generating realistic credit market fluctuations in response to this TFP shock. In the top row, we see that belief dynamics are key drivers of the different behavior of the two models. The decline in investment (top left) is larger in the DE model and moves together with a larger drop in profit forecasts (top right) than in the RE case. So DE beliefs appear critical to quantitatively

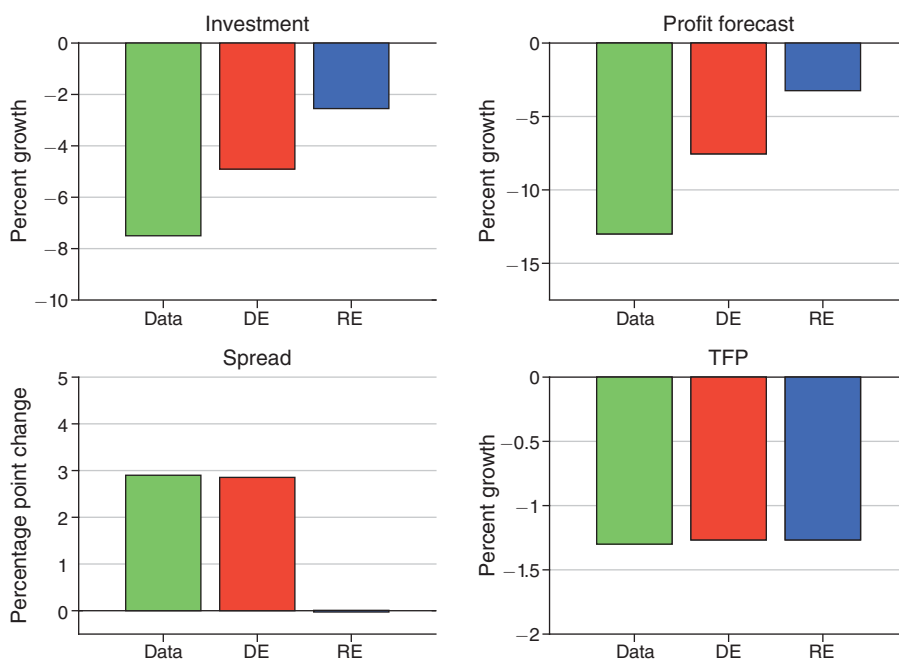


FIGURE 4. THE FINANCIAL CRISIS OF 2007–2009

Notes: Each panel plots a macro series from our crisis decomposition exercise, with the data (green), DE model (red), and RE model (blue) included. The panels plot changes from the precrisis to crisis periods. In the data, all empirical values are averages drawn from the precrisis (2004–2007) or crisis (2008–2009) periods. Spread is the average spread across firms in our Mergent FISD-TRACE-Compustat sample, private nonresidential fixed investment is from NIPA, the macro TFP level is from John Fernald's website, and profit forecasts are the sum of predicted earnings across all firms in our Compustat-IBES guidance data. In the DE model, we choose the TFP growth series in the bottom right panel in order to exactly match the empirical spread values in the bottom left panel. We feed the resulting TFP growth series into the RE model to produce the RE bar in each panel.

account for sudden reversals in macro and financial activity occurring with modest fundamental shocks.

Such reversals are due to fragility built during good times. This fragility entails realistic macro consequences: a large disruption in credit markets and investment, together with a strong reduction in profit forecasts. These macroeconomic changes in the DE model are not identical to those in the data, but the quantitative fit is noteworthy considering the simplicity of our model. Even with the same TFP shock, the RE model produces substantially more modest dynamics. As evident from the impulse response analysis of Section IVB, fragility during good times under DE offers a powerful amplifier of nonlinearities that can help quantitatively account for large crisis episodes even in our simple model.

This exercise shows that DE help account for the salient features of boom-and-bust cycles in macrofinancial aggregates. The realism of the analysis and its fit can be improved by adding ingredients that have likely played a role, such as the housing bubble, intermediary leverage, and the link between household debt and consumption. Boom-and-bust belief dynamics, however, seem a realistic and parsimonious ingredient to generate overexpansion in good times and a sharp financial and real contraction after modest TFP shocks.

VI. Robustness

We conclude by reporting the results of two robustness exercises. First, we explore the robustness of our investment nonlinearity result to a range of alternative parameterizations. Second, we endogenize wages in general equilibrium, which can dampen volatility, and again assess the contemporaneous response of macro investment to TFP shocks.

A. Robustness to Alternative Parameterization

In Supplemental Appendix Figure A.7, we explore the robustness of investment nonlinearity to alternative parameterizations of the model. Starting from the DE parameter estimates in Table 4, a parameterization that we label Baseline, we vary each of the parameters to round higher and lower values, recomputing the sensitivity of investment to a negative shock over the full range of initial conditions. Nonlinearity remains meaningful in each of these robustness checks, with a downward-sloping response indicating more fragility during good times.

B. General Equilibrium

Anticipated procyclical price movements may dampen changes in the anticipated marginal product of capital and hence push against volatility or nonlinearity in investment. To assess the importance of this mechanism, we extend the model to general equilibrium. This exercise poses a Krusell and Smith (1998)–style challenge, and we develop a novel solution technique for overcoming this challenge in Supplemental Appendix A. Although our solution method proves accurate and preserves the nonlinearity of the model, it is highly computationally intensive, with a nested inner loop/outer loop alternation that makes structural estimation of the general equilibrium model infeasible.

For a representative household, we model period utility as $U(C, N) = C - \frac{\omega}{1 + \frac{1}{\lambda}} N^{1 + \frac{1}{\lambda}}$, where C is consumption, the disutility of labor is governed by $\omega > 0$,

and the elasticity of labor supply is given by $\lambda > 0$. See Supplemental Appendix A for a full definition of general equilibrium, involving optimal labor supply and savings decisions by the household, optimal lending choices by a competitive group of lenders, optimal capital investment, labor demand, and risky borrowing decisions by a distribution of nonfinancial firms, and various market clearing and consistency conditions.

One aspect of the equilibrium is straightforward to characterize. The real interest rate is constant and pinned down by the inverse of the household's subjective discount factor $0 < \beta < 1$; that is, $R = \frac{1}{\beta}$. However, as usual in this class of heterogeneous firms models with anticipated macro shocks, general equilibrium presents two computational challenges (Krusell and Smith 1998). In particular, let $\mu(s, k, b)$ be the cross-sectional distribution of micro states $s = (z, \varepsilon_z)$, capital k , and debt b . The macro state is (μ, A, ε_A) . The first challenge is that the macro state (μ, A, ε_A) is intractable because μ is a distribution. The second challenge is that the mapping

$W(\mu, A, \varepsilon_A)$ is a complicated implicit object that must be consistent with the firm-level decisions embedded in market clearing through

$$(19) \quad \left(\frac{W(\mu, A, \varepsilon_A)}{\omega} \right)^\lambda = \int n(s, k, b, \mu, A, \varepsilon_A | W) d\mu(s, k, b),$$

where the left-hand side is the household's closed-form labor supply and the right-hand side reflects labor demand generated by the current cross-sectional distribution of firms μ .

We follow a novel computational approach tailored to our problem and detailed in Supplemental Appendix A. Our approach is to replace the macro state $(\mu_t, A_t, \varepsilon_{A_t})$ with a truncated history of macro shocks $(A_t, A_{t-1}, \dots, A_{t-K})$, nonparametrically storing predictions of the wage W_t given each shock history. We then follow an outer loop/inner loop approach, guessing a wage mapping, solving and simulating the model, and updating the wage predictions until convergence. Our solution technique proves tractable and accurate in practice. We parameterize the model based on the estimated values from Table 4. We further assume a conservative Frisch elasticity of labor supply of $\lambda = 0.5$ and choose β to deliver the same fixed 4 percent annual real interest rate as considered above.

In the general equilibrium model, a positive productivity shock increases perceptions of the future demand for capital and labor and hence future wage growth relative to the partial equilibrium model. These anticipated wage increases dampen investment today. Our general equilibrium solution investigates whether the investment nonlinearities highlighted above under partial equilibrium survive these price movements. Supplemental Appendix Figure A.6, the general equilibrium analog of Figure 2, reports the response of investment to a negative TFP shock for different initial conditions. Endogenous wages moderate the magnitude of the investment response relative to partial equilibrium. But the investment response continues to display substantial nonlinearity under DE (red line): The negative shock exerts a much larger negative impact in good times. In the RE model (blue line), by contrast, the general equilibrium feedback working through wages eliminates the already very slight state dependence of investment almost entirely. In sum, even after allowing for fully flexible wages in general equilibrium, the DE economy proves more fragile than the RE economy in good times and hence more responsive to negative shocks due to the boom-and-bust mechanism created by overreacting beliefs.

VII. Conclusion

The financial crisis of 2008 renewed economists' interest in financial instability. One key challenge is to understand where such instability comes from. We showed that nonrational beliefs, and in particular overreaction to news, can generate realistic credit cycles without relying on financial shocks. These boom-and-bust dynamics exhibit predictability in line with the evidence on cyclical movements in credit spreads (López-Salido, Stein, and Zakrajšek 2017) and on large financial crises (Greenwood et al. 2022). Critically, our results are obtained in a neoclassical model in which a single new parameter, the degree of belief overreaction, is estimated using microdata on managers' errors in forecasts of the earnings growth of their

firms. Realistic micro-level belief distortions can, once aggregated, generate realistic credit cycles with financial overexpansion in good times, fragility, and sharp reversals as small adverse news arrives.

Future work can enrich our approach. A key factor is the role of financial intermediaries, which recent work has already begun to investigate (Maxted 2023; Krishnamurthy and Li 2025). Another important aspect is household debt, which has been shown to be a key determinant of drops in aggregate demand during financial tightenings (Mian, Sufi, and Verner 2017; Mian and Sufi 2009). Prolonged asset price bubbles may also be important, especially to account for large crises (Greenwood et al. 2022). These factors open exciting avenues for studying the transmission of beliefs to the real economy. As an initial step, we show that realistic departures from rationality disciplined by expectations data can be introduced into standard macroeconomic models and significantly improve their explanatory power.

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